

Rui Bian, Ph.D.

FOUNDING AI & DATA ENGINEER | AI, DATA PLATFORMS, BACKEND SOFTWARE

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PROFESSIONAL SUMMARY

Founding AI and Data Engineer with 14+ years across production software, data platforms, applied AI/ML, and doctoral systems research. Builds reliable AI-powered products end to end: backend APIs, data ingestion, workflow orchestration, model operationalization, observability, and operator experiences. Previously owned 32 production systems serving 30+ school districts, with an emphasis on fault isolation, idempotent recovery, explainable automation, and privacy by architecture.

TECHNICAL SKILLS

AI & ML Systems	LangGraph, local LLMs, OpenAI API, Gemini API, scikit-learn, XGBoost, SHAP, MLflow
Backend & Software	Python, Go, TypeScript, FastAPI, Flask, REST APIs, Pydantic, Docker, pytest, CI/CD
Data Platforms	SQL Server, PostgreSQL, MongoDB, SQLAlchemy, pandas, NumPy, Apache Spark, Microsoft Fabric, ETL
Automation & Analytics	Playwright, Selenium, asyncio, Power BI, DAX, Streamlit, Plotly, Cloudflare Workers, AWS, Azure

EXPERIENCE

Founding AI and Data Engineer

Aug 2026 - Present

Big Shot Pictures

- Lead 0-to-1 design and implementation of core AI and data infrastructure for data-driven creative and business operations.
- Build scalable pipelines, storage, analytics workflows, AI-driven features, and agentic frameworks for production workflows and internal capabilities.
- Partner with cross-functional leadership to translate business vision into robust, enterprise-grade engineering strategy and delivery.

Lead Data Scientist

Dec 2022 - Jun 2026

Expatiate Communications

- Architected, shipped, and owned 32 production AI, data, and automation systems serving 30+ California school districts; reduced new-district onboarding from weeks of engineering to hours of configuration.
- Built a LangGraph orchestration platform with conditional routing, parallel fan-out, local LLM diagnostics, and node-level fault isolation; reduced recurring pipeline runtime by 75-90 percent.
- Operationalized six-domain predictive decision support with per-district model selection, explainability, idempotent batch scoring, and graceful LLM provider fallback across 20+ districts.
- Automated compliance workflows across 21 districts using SQL, MFA-aware Playwright, PDF parsing, auditable deadline rules, and Power BI; supported zero missed IEP deadlines after deployment.
- Delivered 150+ dashboards and operator-first automation, including weekly reporting across 18 school sites, so non-technical teams could run recurring workflows without engineering support.

Ph.D. Researcher - Computer Engineering

2015 - 2022

University of Delaware

- Built Python and AWS measurement systems processing millions of internet probes for proxy ecosystems, traffic integrity, and global routing analysis.
- Analyzed 436,000+ open proxies and developed detection and classification methods for transparent proxies and anycast routing anomalies.
- Published at IEEE INFOCOM, Computer Networks, and ACM SIGCOMM CCR; served as a reviewer and teaching assistant in networking and cybersecurity.

SELECTED PLATFORM EVIDENCE

AI Data Workflow Orchestration Platform | LangGraph, Ollama, Python

Gate-node routing, four-stage parallel execution, local failure diagnosis, audit logs, and rerun-safe boundaries across 30+ district workflows and 100K+ records per run.

Concurrent SIS Data Ingestion Service | Go, MongoDB, REST

Goroutine fan-out across 12 dataset types with per-school timeout/authentication isolation, startup validation, health checks, dynamic school discovery, and idempotent reloads.

SELECTED API & PRODUCT WORK

Adaptive Creative Analysis API | FastAPI, Pydantic, SQLAlchemy, Docker

Schema-validated creative analysis with explainable model-tier routing, bounded retries, deterministic fallback, persisted lifecycle state, provider metadata, health checks, and automated tests.

California School Explorer | React, TypeScript, Python, PostgreSQL, Cloudflare

Open-source public-data product with 9,946 school profiles, 3.9M+ canonical facts, subgroup and context-aware comparison, source notes, suppression handling, and reproducible processing.

Free Image Tools | JavaScript, Canvas API, Cloudflare

Free, open-source, privacy-first browser utilities for compression, conversion, resizing, PDF workflows, metadata removal, batch processing, and optional AI alt text.

EDUCATION

Ph.D. in Computer Engineering
University of Delaware | GPA 3.96/4.0

2015 - 2022

M.S. in Engineering
University of Science and Technology of China

2012 - 2015

B.S. in Engineering
University of Science and Technology of China

2008 - 2012

CERTIFICATIONS & CONTINUOUS LEARNING

DataCamp (2026, valid through 2028): AI Engineer for Developers Associate; AI Engineer for Data Scientists Associate; Data Scientist Associate; Data Engineer Associate.

Google Cybersecurity Professional Certificate, Coursera, 2023.

SELECTED PUBLICATIONS & PATENT

- "Silent Observers Make a Difference: A Large-scale Analysis of Transparent Proxies on the Internet." IEEE INFOCOM, 2024.
- "Shining a Light on Dark Places: A Comprehensive Analysis of Open Proxy Ecosystem." Computer Networks, 2022.
- "Towards Passive Analysis of Anycast in Global Routing: Unintended Impact of Remote Peering." ACM SIGCOMM CCR, 2019.
- Patent CN104614936B: "Manufacturing Method of Micro Lens."

ADDITIONAL EXPERIENCE & RECOGNITION

- Master research in engineering at USTC and a research internship at the Changchun Institute of Optics, Fine Mechanics and Physics, Chinese Academy of Sciences.
- SANS CyberStart Scholarship - ranked 44th of 3,935 participants (top 1 percent).
- Reviewer and TPC service for IEEE INFOCOM, IEEE/IFIP DSN, IEEE TNSE, Computer Networks, and related systems venues.